Task result overview

Task information

Software version: 1.2.10

Generated UTC: 2026-10-11T14:22:09Z

Task name: reference_network_report_demo

Analysis duration: 2 h

Time step: 60 s

Thermal network: reference_11_node_gnss_receiver

Outputs from this run

scenario

Main findings

Mean beta angle: 19.46 deg

Eclipse fraction: 28.93 %

Response-node temperature P95-P5: 13.69 K

Simulated code-bias P95-P5: 2.685 m

Power-to-temperature peak time lag 2201.2 s (1 eligible cycle).

Maximum energy closure: 2.001e-09 J

Orbit heating and nodal response

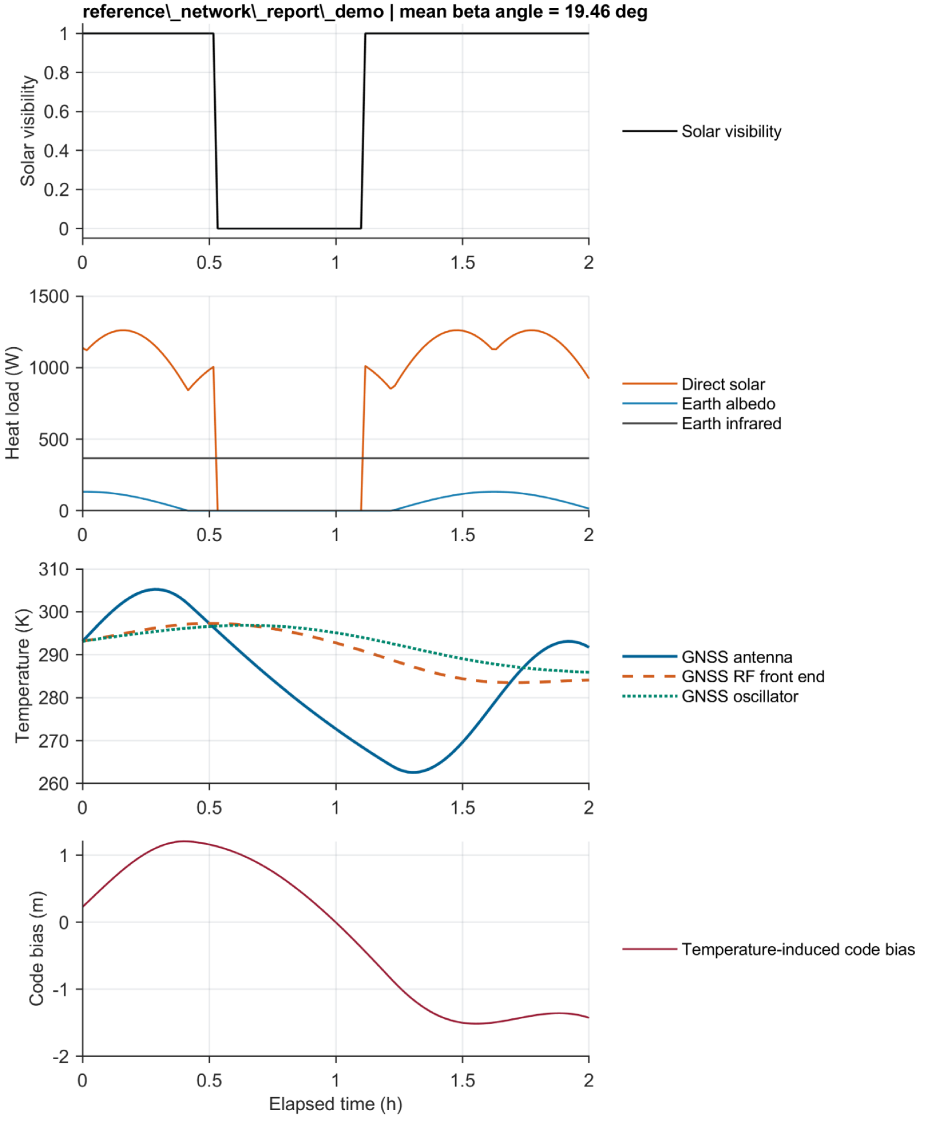
Mean beta angle: 19.46 deg; eclipse fraction: 28.93%.

Response-node temperature P95-P5: 13.69 K; assumed temperature-induced code-bias P95-P5: 2.685 m.

Power-to-temperature peak time lag 2201.2 s (1 eligible cycle).

2 illumination transitions occurred over this run.

No warmup was used; initial-temperature transients may affect spans and lag.



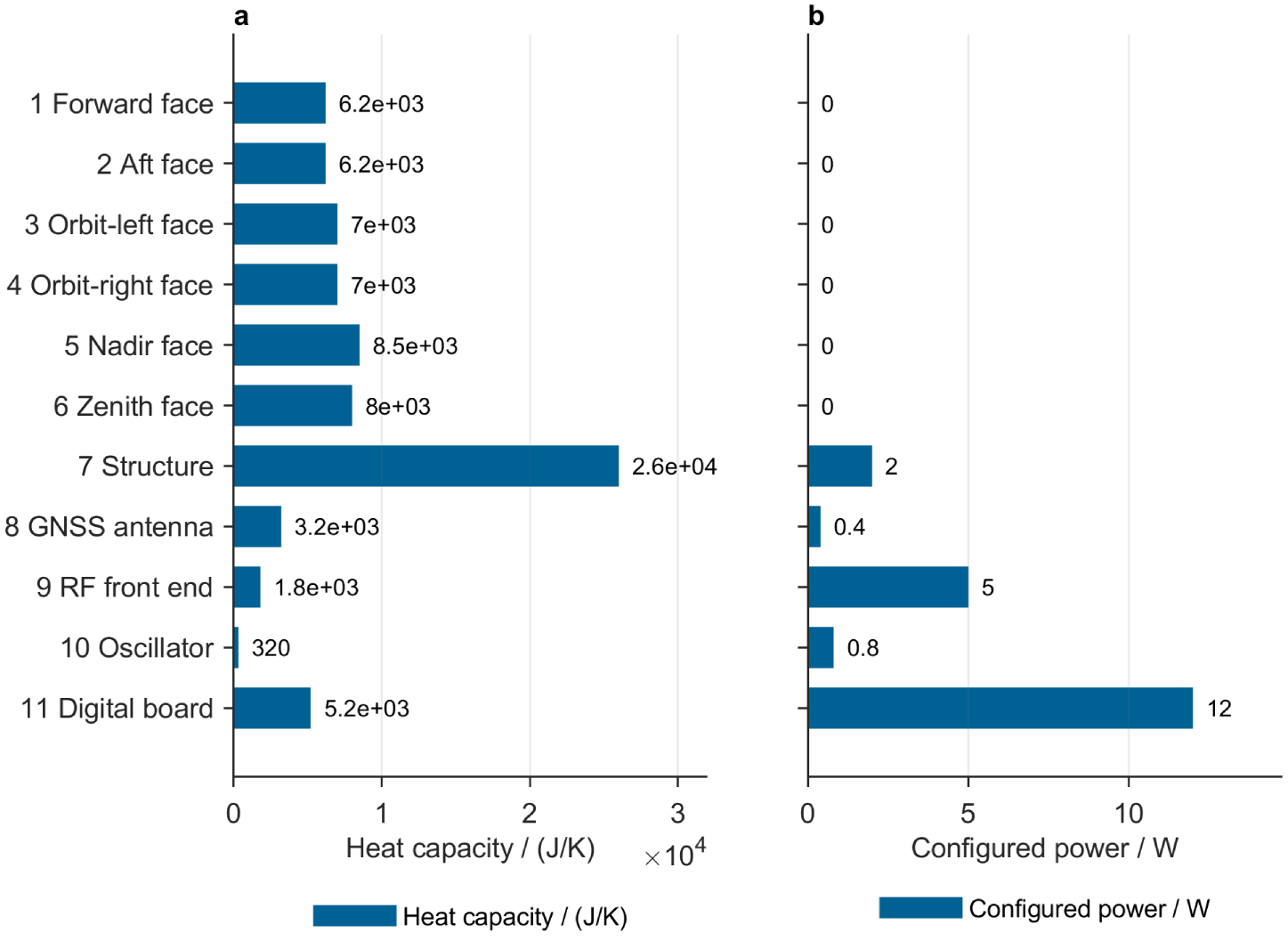
Heat capacity and internal dissipation

11 nodes; total capacity 79420 J/K; configured internal power 20.2 W.

Spacecraft structure has the largest capacity: 26000 J/K (32.7% of total).

Receiver digital board has the highest configured power: 12 W.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



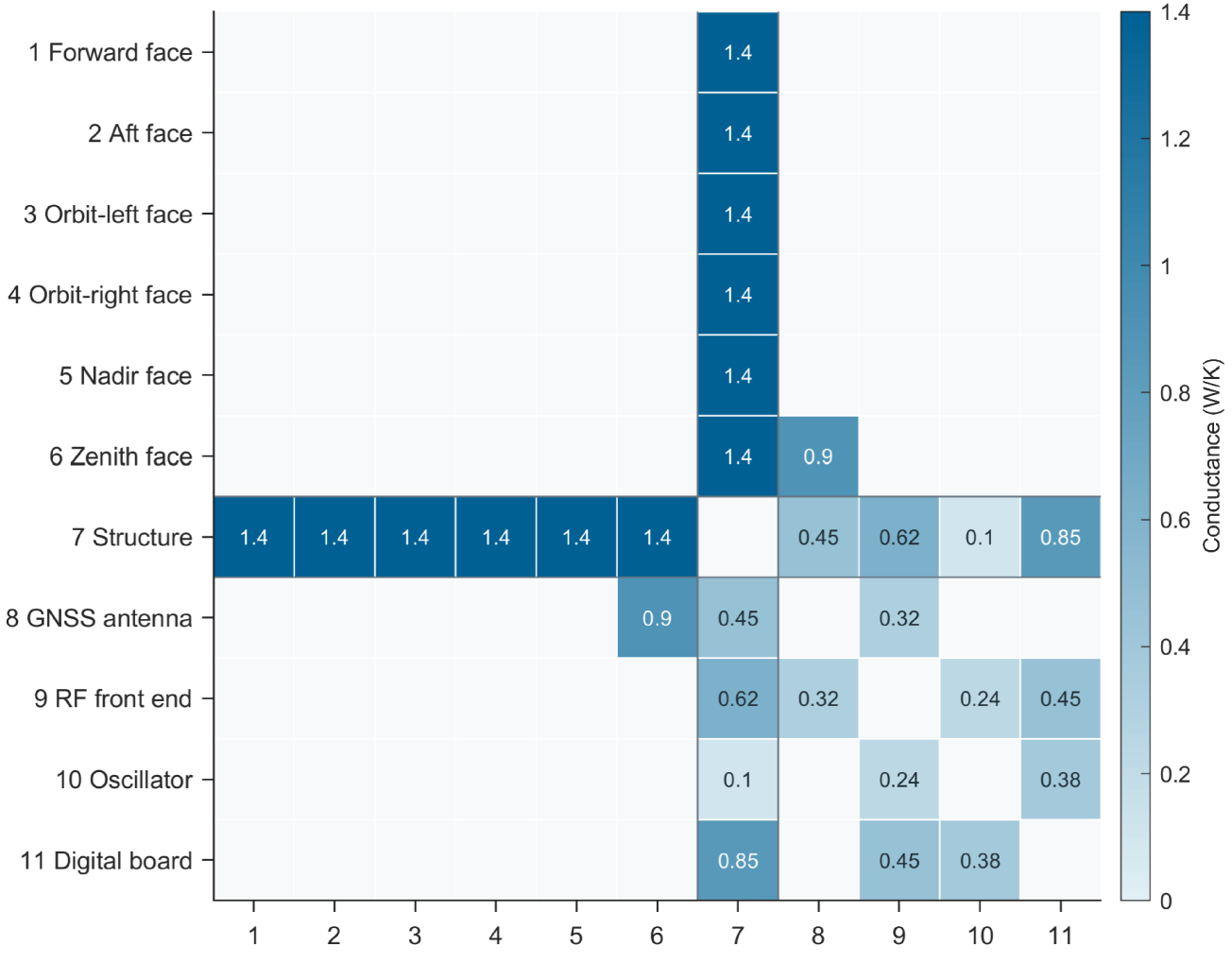
Thermal links and conductance

15 conductive links; Spacecraft structure has the most connections (10).

First 11/11 nodes shown; filled cells and values give conductance. Blank cells have no direct conductive link.

The symmetric matrix shows each reciprocal link twice. Conductances are listed in the link CSV.

Among links connected to GNSS RF front end, the largest absolute mean net transfer is GNSS RF front end to Spacecraft structure at 7.190 W. Total mean net conduction from the response node is 7.260 W (positive outward). This identifies an active path using heat flow rather than link count.

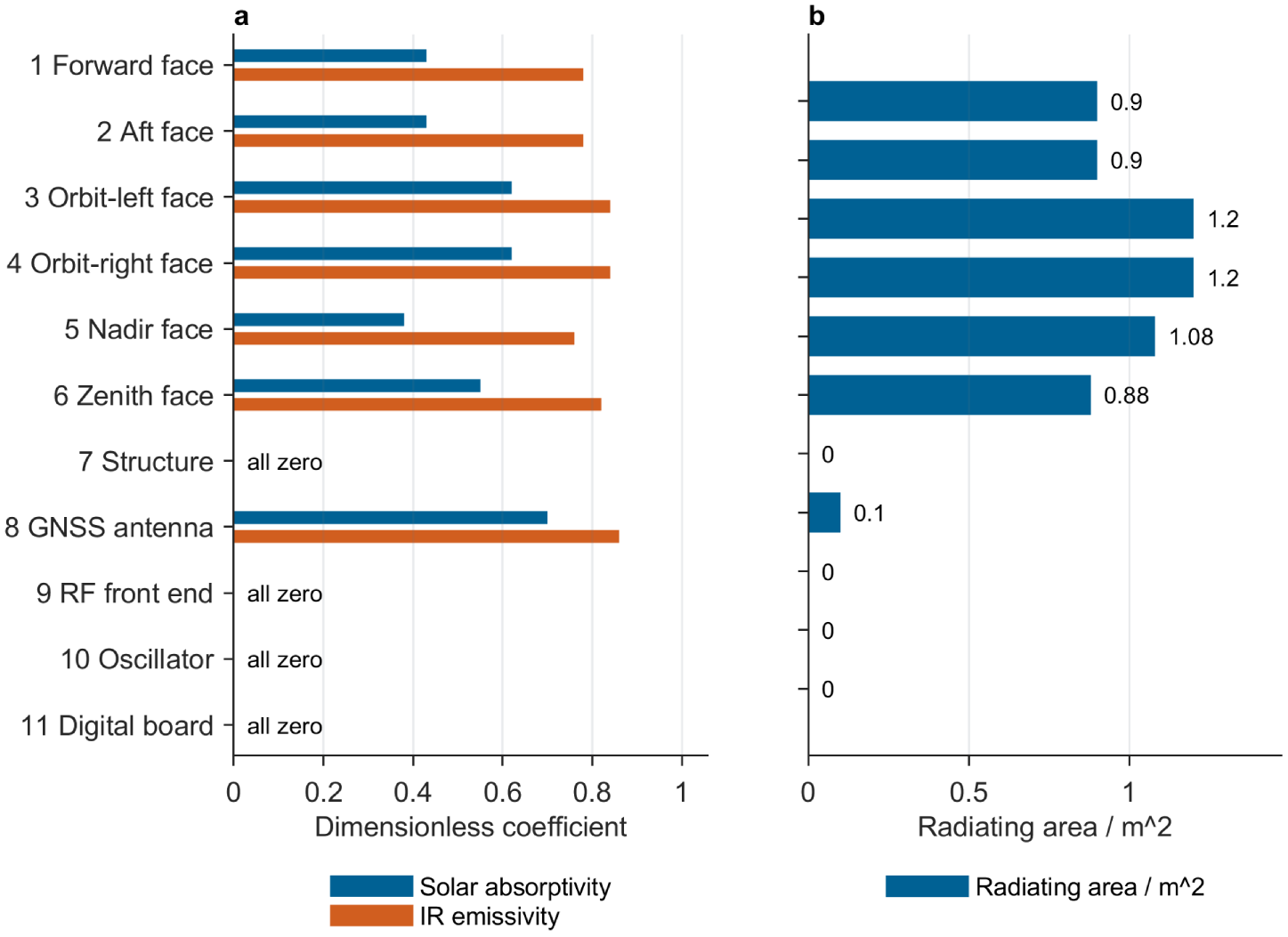


Optical properties and radiating area

Total external radiating area: 6.26 m^2. Zero values are explicitly labelled.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Spacecraft structure, GNSS RF front end, GNSS oscillator, Receiver digital board have zero projected and external radiating areas and exchange heat through conductive links.



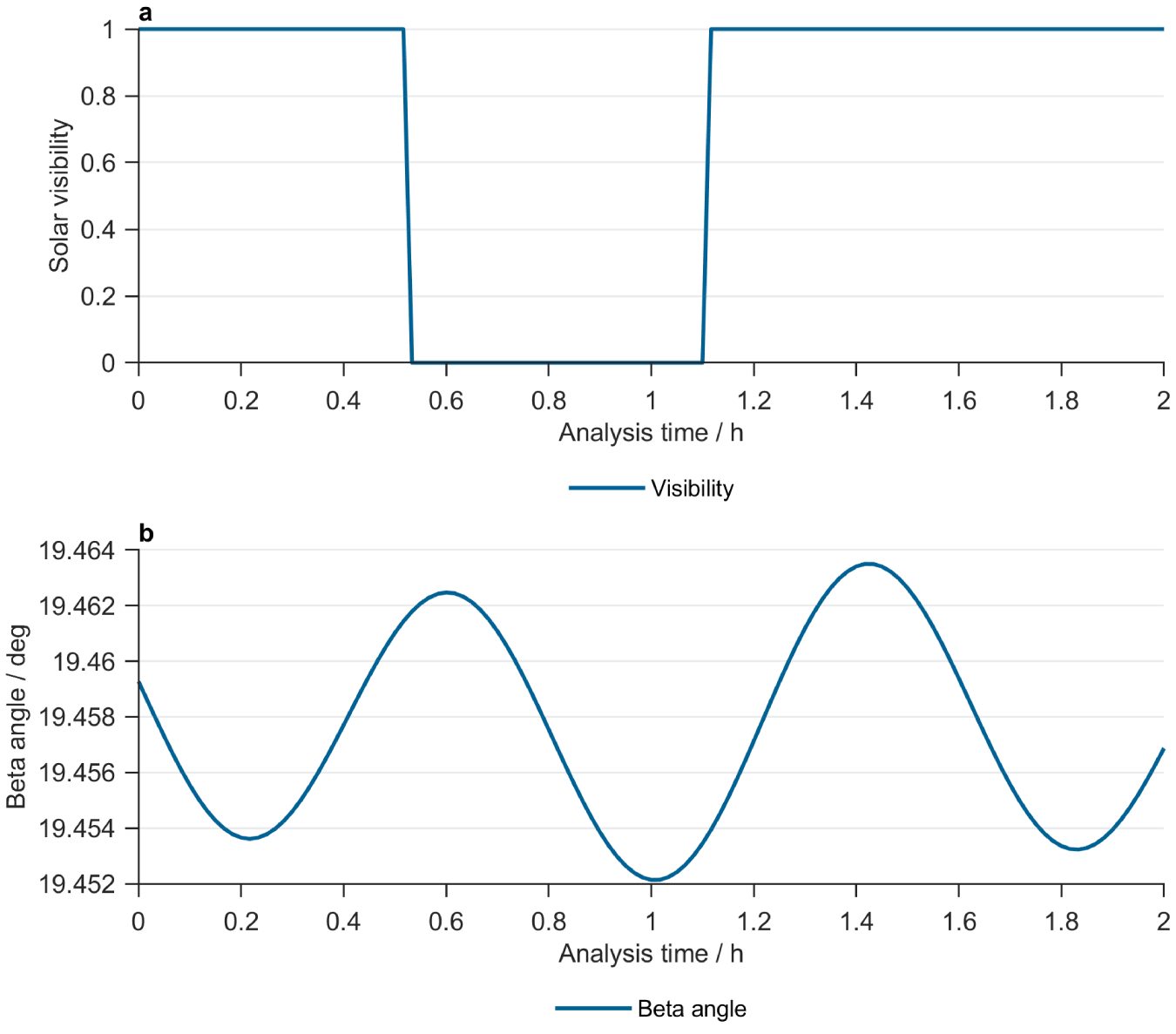
Illumination and eclipse structure

Trapezoidal time weighting estimates partial/full eclipse over 29.17% of the window.

Solar visibility ranges from 0.000 to 1.000; orbital period 96.69 min.

Eclipse uses visibility below 0.999. Time-weighted and epoch-count fractions can differ.

1 eclipse entries and 1 exits; beta angle 19.452 to 19.463 deg.



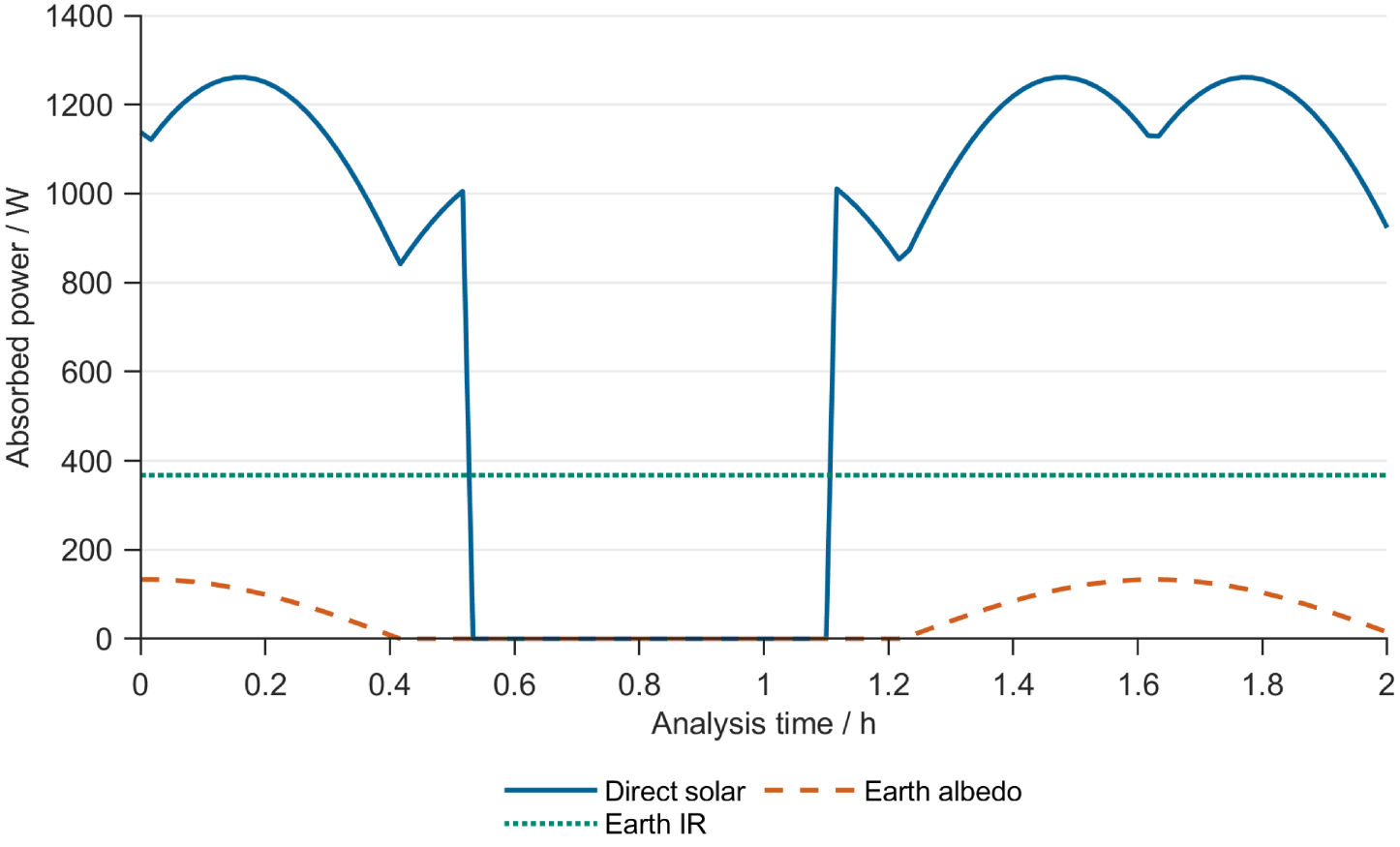
Environmental heat inputs

Time-mean absorbed power: solar 795.415 W; albedo 51.963 W; Earth IR 367.422 W.

Curves sum absorbed power over the network, including area and absorption properties.

Solar absorbed power 0.000 to 1261.402 W; Earth IR 367.422 to 367.422 W.

Time-integrated external heat shares are 65.5% direct solar, 4.3% albedo and 30.2% Earth infrared. direct solar contributes most to external absorption; these are heat-input shares, not temperature-response attribution.



Nodal temperatures and extrema

Window maximum 315.304 K (42.15 deg C, node 6); minimum 220.361 K (-52.79 deg C, node 3).

Largest full peak-to-peak range 80.070 K (node 2), distinct from a temperature P95-P5 difference.

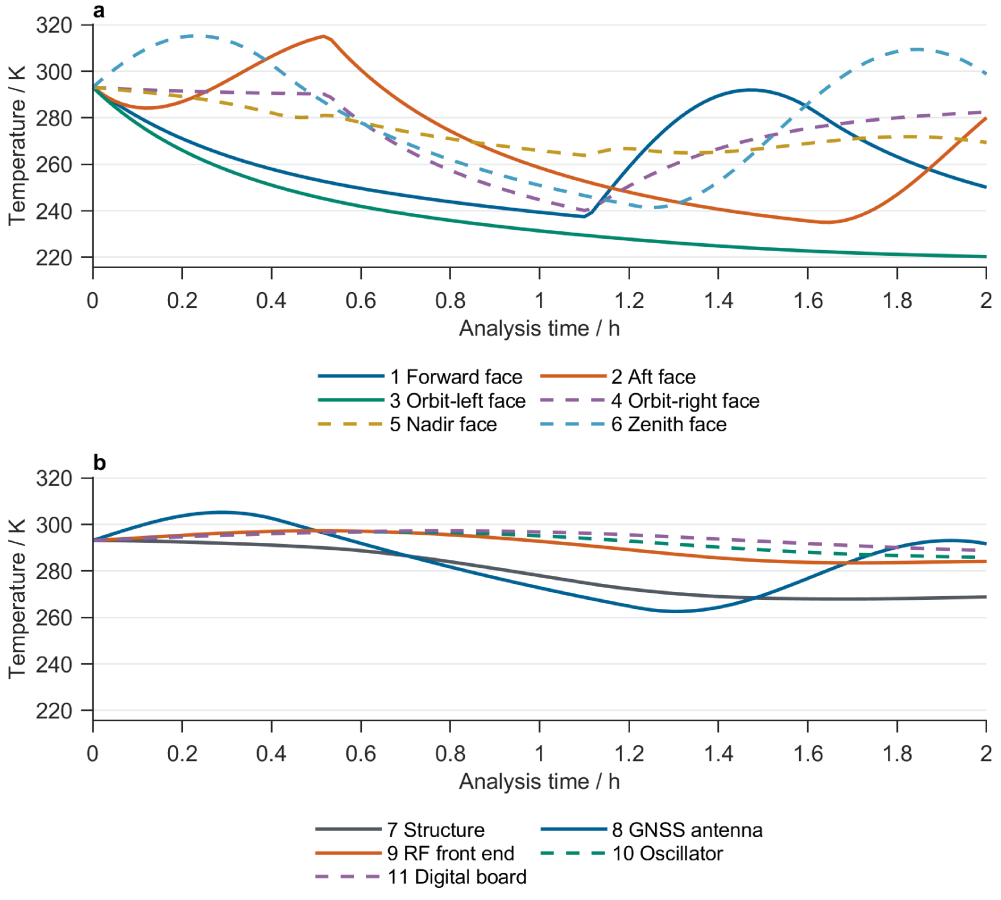
Hottest node: Zenith face; coldest node: Orbit-left face.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Power-to-temperature peak time lag 2201.2 s (1 eligible cycle).

Temperature level and variation describe different demands: the maximum is at Zenith face (315.30 K), the minimum at Orbit-left face (220.36 K), and the largest full range at Aft face (80.07 K).

GNSS RF front end has a full range of 13.805 K, 17.2% of the largest nodal range. Assess response-node stability directly rather than substituting exterior or network-wide extremes.

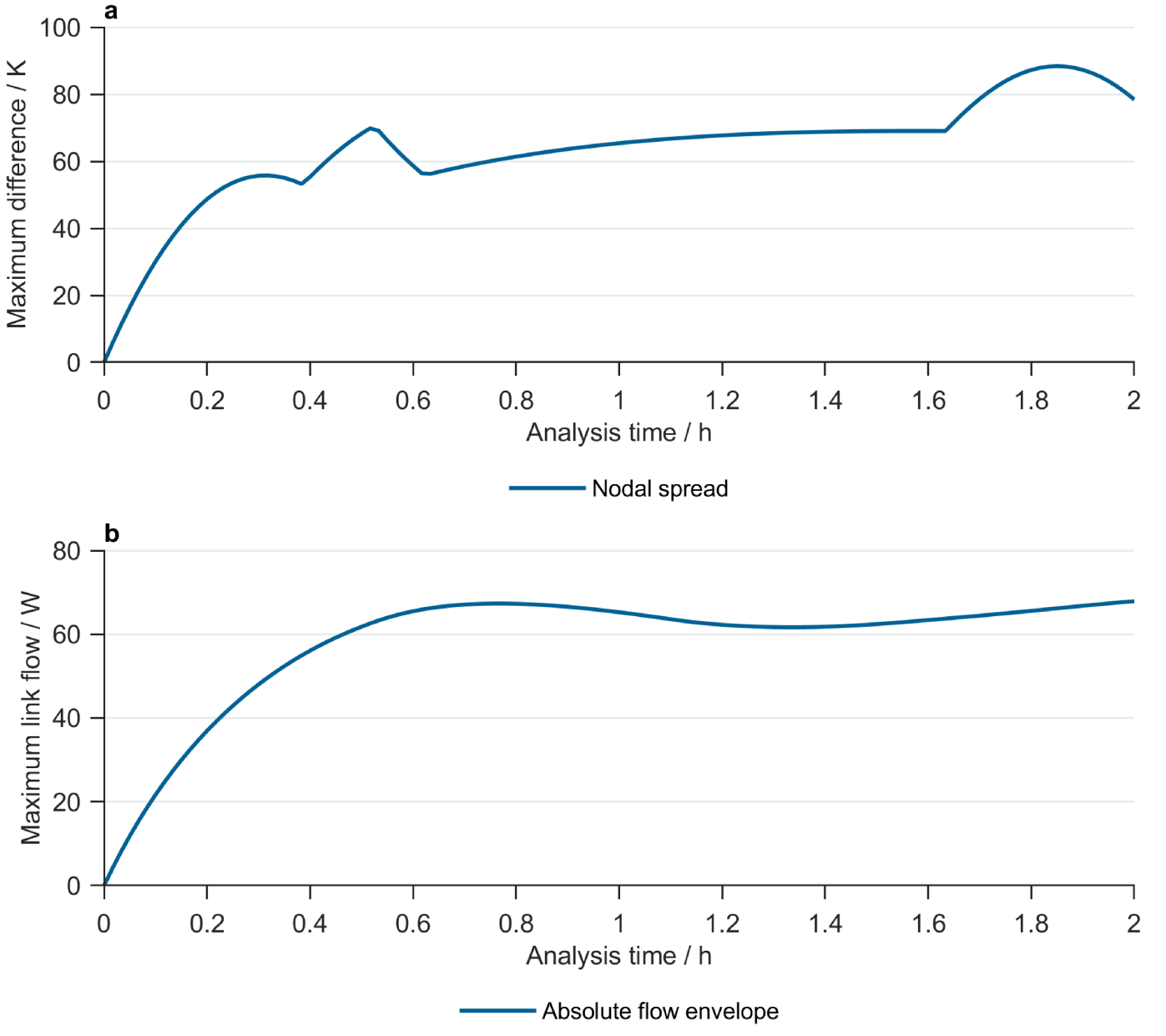


Temperature differences and link heat flow

Maximum simultaneous nodal spread 88.414 K; maximum absolute single-link flow 67.888 W.

Panels show epoch-wise temperature spread and the largest absolute link heat flow.

The envelope takes the largest value across 15 conductive links at each epoch.



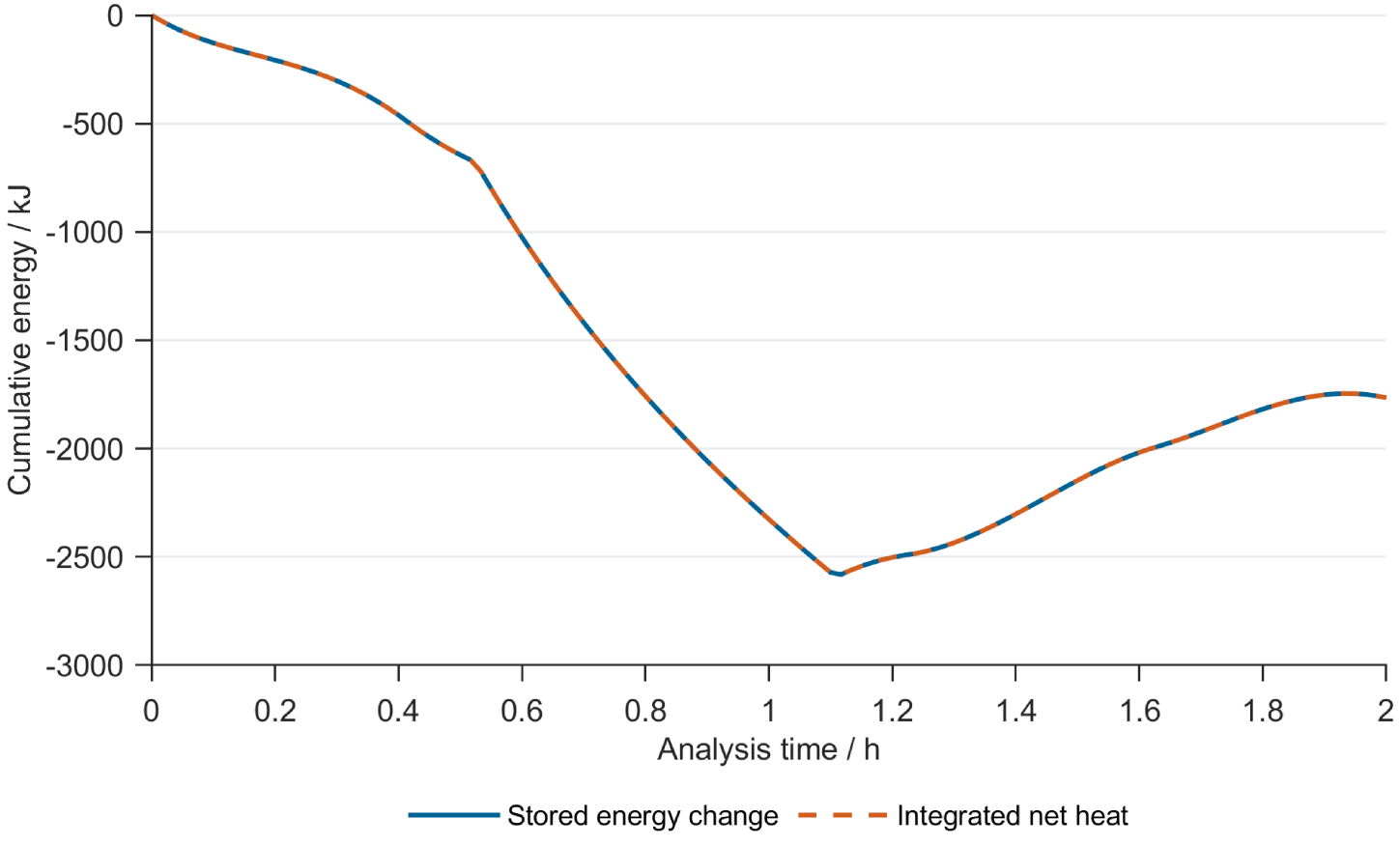
Stored energy and energy balance

After rebasing both at the analysis start, maximum cumulative closure is 6.0536e-09 J.

Stored energy is the sum of capacity times temperature change, relative to analysis start.

Balance: stored energy = external heat + internal heat - radiation + boundary heat.

End-of-window stored-energy change: -1765.6 kJ.



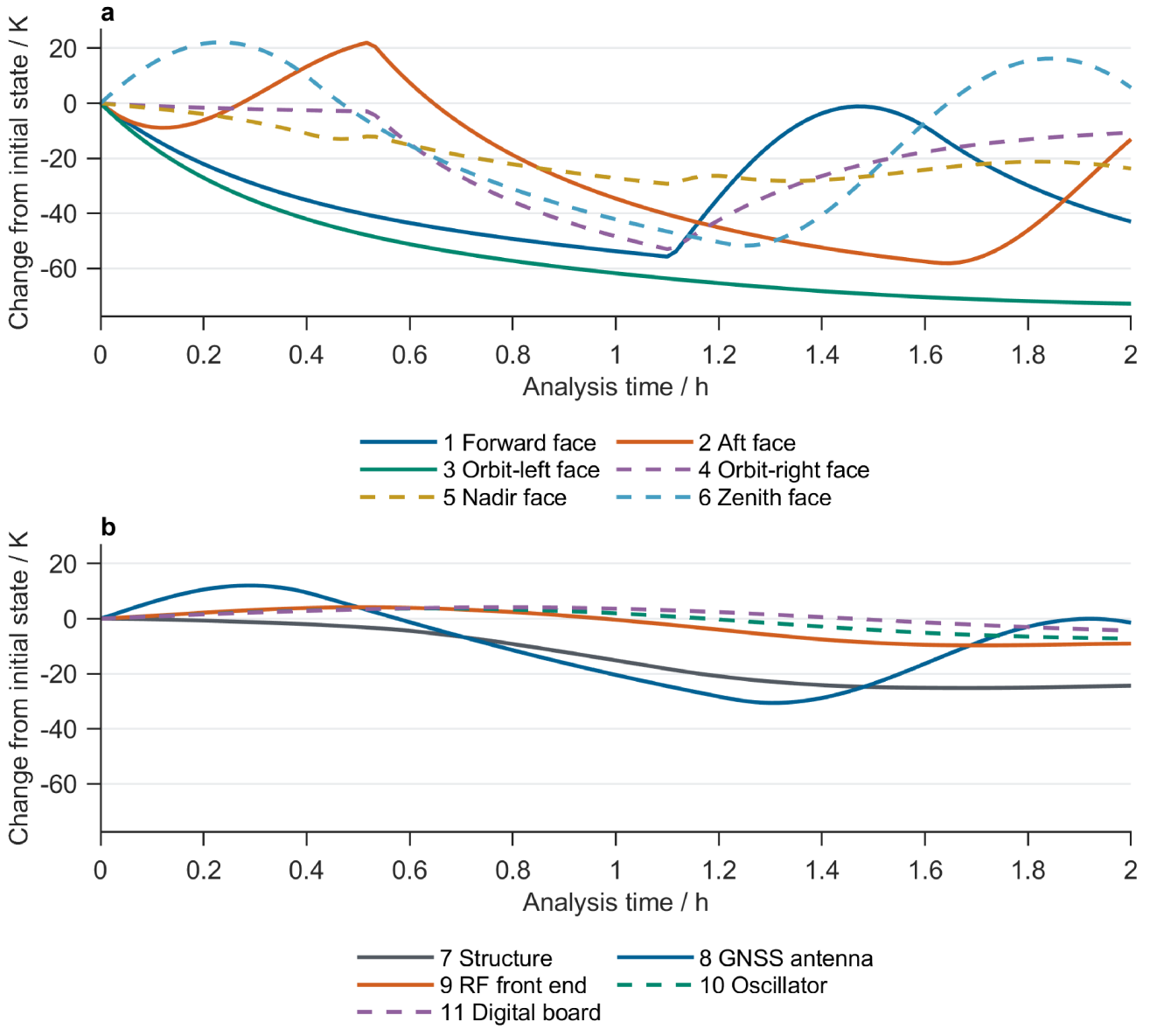
Initial state and temperature drift

Fixed warmup: 0 orbits. Periodic-state convergence was not tested.

Largest absolute end-minus-start temperature change 72.789 K.

Endpoints are 1.241 orbits apart; curves are relative to analysis-start temperatures.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



Orbit and effective thermal conductance: 36-case study

The 36 cases combine 400, 600 and 800 km altitudes, target beta angles of 0, 30, 60 and 75 deg, and effective structure-to-RF thermal conductances of 0.31, 0.62 and 1.24 W/K. Other equipment parameters remain fixed.

Each case is initialized to a periodic temperature difference of at most 0.001 K, then analyzed over six orbits with 360 output intervals per orbit. Phase delay is referenced to total external power absorbed by all nodes.

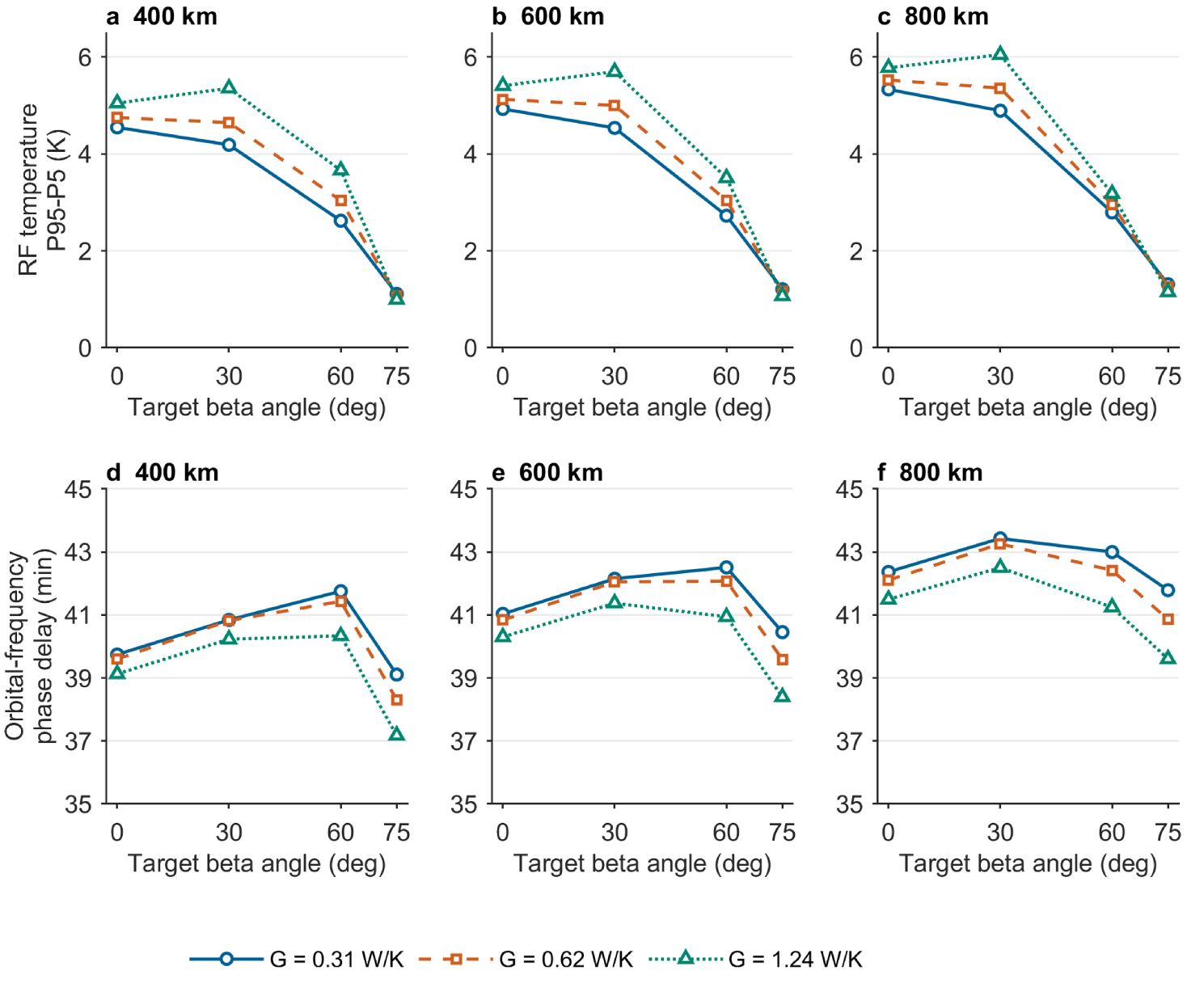
RF front-end temperature P95-P5 ranges from 0.995 to 6.048 K.

Increasing effective thermal conductance from 0.31 to 1.24 W/K increases P95-P5 by 8.4-39.7% in eclipsing cases and decreases it by 10.5-12.1% under continuous sunlight.

Across the 12 altitude-beta pairs, orbital-frequency phase delay decreases by 36.8-131.0 s (1.5-5.2%). Conductance affects temperature fluctuations differently across illumination conditions, whereas phase delay decreases in all tested pairs.

Three prespecified cases are independently recomputed with 720 intervals per orbit; maximum changes are 0.0005757 K in P95-P5 and 0.1269 s in phase delay.

Orbit and effective thermal conductance: 36-case study



All-node statistics

Temperature: K; capacity: J/K; configured power: W. Means are time-weighted.

1: Forward face

Min 237.465 | Mean 262.605 | Max 293.150 | Full range 55.685

Capacity 6200 | Internal power 0 | End minus start -42.980 K

2: Aft face

Min 235.058 | Mean 268.154 | Max 315.128 | Full range 80.070

Capacity 6200 | Internal power 0 | End minus start -13.050 K

3: Orbit-left face

Min 220.361 | Mean 237.700 | Max 293.150 | Full range 72.789

Capacity 7000 | Internal power 0 | End minus start -72.789 K

4: Orbit-right face

Min 240.166 | Mean 272.306 | Max 293.150 | Full range 52.984

Capacity 7000 | Internal power 0 | End minus start -10.680 K

5: Nadir face

Min 263.868 | Mean 273.995 | Max 293.150 | Full range 29.282

Capacity 8500 | Internal power 0 | End minus start -23.743 K

6: Zenith face

Min 241.432 | Mean 279.932 | Max 315.304 | Full range 73.872

Capacity 8000 | Internal power 0 | End minus start 5.658 K

All-node statistics (continued)

7: Spacecraft structure

Min 267.987 | Mean 279.282 | Max 293.150 | Full range 25.163

Capacity 26000 | Internal power 2 | End minus start -24.297 K

8: GNSS antenna

Min 262.602 | Mean 284.306 | Max 305.227 | Full range 42.626

Capacity 3200 | Internal power 0.4 | End minus start -1.424 K

9: GNSS RF front end

Min 283.497 | Mean 290.880 | Max 297.302 | Full range 13.805

Capacity 1800 | Internal power 5 | End minus start -9.039 K

10: GNSS oscillator

Min 285.918 | Mean 292.700 | Max 296.859 | Full range 10.941

Capacity 320 | Internal power 0.8 | End minus start -7.232 K

11: Receiver digital board

Min 288.816 | Mean 294.430 | Max 297.306 | Full range 8.490

Capacity 5200 | Internal power 12 | End minus start -4.334 K

Inputs and result details

Scenario name: reference_network_report_demo

Duration: 7200 s

Time step: 60 s

Fixed warmup setting (non-convergence mode): 0 orbit

Node count: 11 node

Conductance range: [0, 1.4] W/K

Internal power range: [0, 12] W

Start epoch UTC: 2024-03-20 12:00:00

Altitude: 600000 m

Inclination: 97.6 deg

Attitude mode: nadir

Trajectory coordinate system: Geocentric equatorial axes (built-in approximate orbit model)

Telemetry comparison: Not run (no telemetry data imported for this task)

Computation checks and audit record

Numerical and configuration checks

Finite scenario temperatures: Pass

Telemetry comparison: Not run (no telemetry data imported for this task)

Assessment scope

This report evaluates the configured thermal simulation and numerical checks. Device temperature-limit compliance is outside this assessment.

Audit files

MAT results, metric CSV, Markdown, manifest, and figures accompany this PDF.

Software version: 1.2.10

Task conclusions

1. Prioritize the high-temperature level at Zenith face, low-temperature level at Orbit-left face, and variation at Aft face in design review: extrema 315.30/220.36 K and largest range 80.07 K.

2. Prioritize direct solar in environmental-input review: it supplies 65.5% of absorbed external heat. Interpret temperature response together with the heat-transfer paths and orbital phase.

3. The response component GNSS RF front end spans 283.497 to 297.302 K, with a full range of 13.805 K. Use these component values separately from spacecraft extrema.

4. Prioritize the GNSS RF front end / Spacecraft structure contact in conductance review: mean net transfer is 7.190 W. This ranking uses observed model heat flow, not a conductance-perturbation experiment.